Sahaj Bhadja

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Summary

Software Engineer with 2+ years building customer-facing **Python** and **React** products, marketplace booking APIs, **search** and retrieval, and production AI workflows, with a high bar for **automated testing**, monitoring, and data-driven decisions.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured asynchronous **Python** AI orchestration with FastAPI workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Tuned **query and indexing** patterns on a multi-tenant platform for 500+ users, cutting dashboard latency from 1.2s to 350ms, and shipped validated integration pipelines through CI/CD at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a customer-facing **React** and Flask grading product on AWS for 150+ instructors and 10K+ daily requests, designing frontend-to-backend upload, rubric, and grading-history workflows end to end.
- Integrated GPT-4o, Claude, and Qwen-VL behind a switchable **AI model layer** with OCR fallbacks and per-call cost tracking, reaching 90%+ grading consistency and 60% faster instructor feedback.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built end-to-end Python **ML pipelines** turning 100K+ trajectory logs and map features into training data for a PyTorch Decision Transformer, improving long-horizon planning 18% in closed-loop simulation.
- Ran **model evaluations** comparing a new attention encoder against three baselines in Hydra and Weights & Biases experiments, improving closed-loop planning stability 15% and documenting failure modes.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **Django and React** platform with RESTful APIs and real-time WebSocket updates for resource allocation, customers, billing, and leads, syncing 2M+ records across locations with 33% fewer sync conflicts.
- Gated releases on **automated** unit, integration, and end-to-end tests in Jenkins, monitored with Sentry and Kibana, and used **data-driven** analytics to cut resource wastage 42% and lift lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built backend APIs for a **two-sided marketplace** connecting households with cleaning, repair, and child-care providers, covering bookings, time slots, pricing, and coupons across 50K+ records with sub-200ms responses.
- Supported **React Native** Android and iOS clients and a React TypeScript web dashboard with Firebase real-time booking status updates, improving confirmation speed 40% and cross-platform compatibility 15%.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, HTML, CSS
Backend & Search: Django, Flask, FastAPI, REST APIs, Microservices, FAISS, Pinecone, Semantic Search, Indexing
Frontend & Mobile: React, React Native, Next.js, TailwindCSS, WebSockets
Data & ML: PostgreSQL, MySQL, Firebase, PyTorch, LLM Systems, Weights & Biases, Spark, Google Analytics
Quality & Delivery: PyTest, Jest, Selenium, Unit Testing, Integration Testing, CI/CD, Jenkins, Sentry, Docker, AWS, Git

Projects

Geo-Aware Provider Search Ranking | *Python, Django, Elasticsearch, PostgreSQL, React, PyTest*

- Designed a Django service that ranks care providers in **Elasticsearch** with distance decay, availability filters, and review signals, plus an offline **relevance** evaluation harness comparing ranking variants before A/B rollout.

Preference-Driven Match Profiles | *Python, Django REST, React Native, PostgreSQL, PyTest*

- Designed a **React Native** intake flow and Django REST API capturing structured client needs as progressive questions, storing them as ranking features and replaying them to providers to raise **match confidence**.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built a **semantic search** platform with transformer embeddings, FAISS, and Pinecone over 50K+ documents, using incremental indexing and tuning to cut retrieval latency 40% while sustaining 95% query accuracy.